BANTAY–BAHA

Early warning for flood-linked disease: which provinces will see a surge after a typhoon, and which of them will be cut off from a hospital.

46%
lower forecast error than the seasonal baseline

7 / 10
top-ranked provinces surged after Typhoon Ompong

100
ferry routes a single typhoon can close, out of 108

TYPHOONS BRING FLOODS. FLOODS BRING DISEASE.

8–9

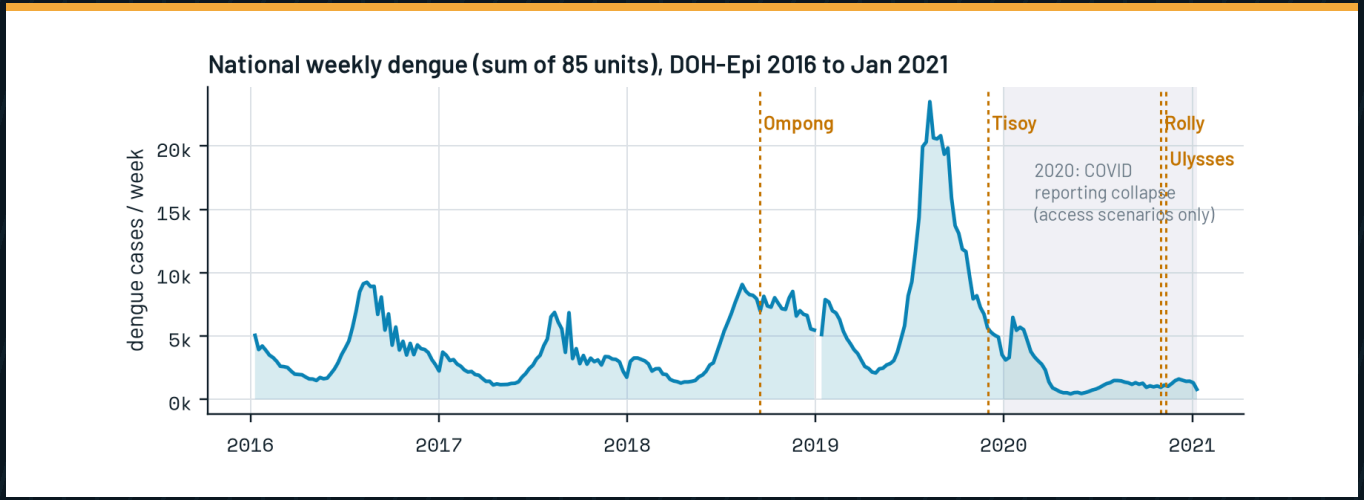
tropical cyclones cross the Philippines every year, mostly from July to October (PAGASA)

2,089

leptospirosis cases and 162 deaths after Typhoon Ondoy in 2009, five times the national lepto deaths of 2008 (Balbuena et al., 2010)

146k

dengue cases from January to July 2019, when DOH declared a national epidemic



WHO IS HIT HARDEST

Floodwater spreads leptospirosis within days; standing water breeds the mosquitoes behind dengue for weeks after. The same storm closes roads and grounds ferries, so the most exposed communities are often the ones that **cannot reach a hospital**.

THE RESPONSE ARRIVES AFTER THE SURGE

A typhoon sets off a sequence. Each step is predictable from data that already exist, yet today's response starts only at the last one.

DAY 0
LANDFALL
Gale-force winds; ferries suspended; roads blocked by landslides and debris

DAYS 1-7
FLOODING
Rivers carry upstream rain to the lowlands; towns at river mouths flood after the rain stops

WEEKS 1-2
LEPTOSPIROSIS
Risk from rainfall peaks about two weeks later and runs through flooding (Matsushita et al., 2018)

WEEKS 2-8
DENGUE
Breeding sites multiply; cases rise province by province

TODAY
RESPONSE
Prophylaxis, vector control and medical teams go out after cases rise; isolated places are reached last

What is missing is a tool that looks at **this** storm and says, before the roads close, **where** disease will rise and **who** will be cut off.

DENGUE FORECASTS EXIST
Climate-driven, province-level (Buczak et al., 2014), but not for a specific storm and without honest ranges

ACCESS STUDIES EXIST
Flood access loss for one city, Metro Manila (Tariverdi et al., 2023), with no health forecast and no ferries

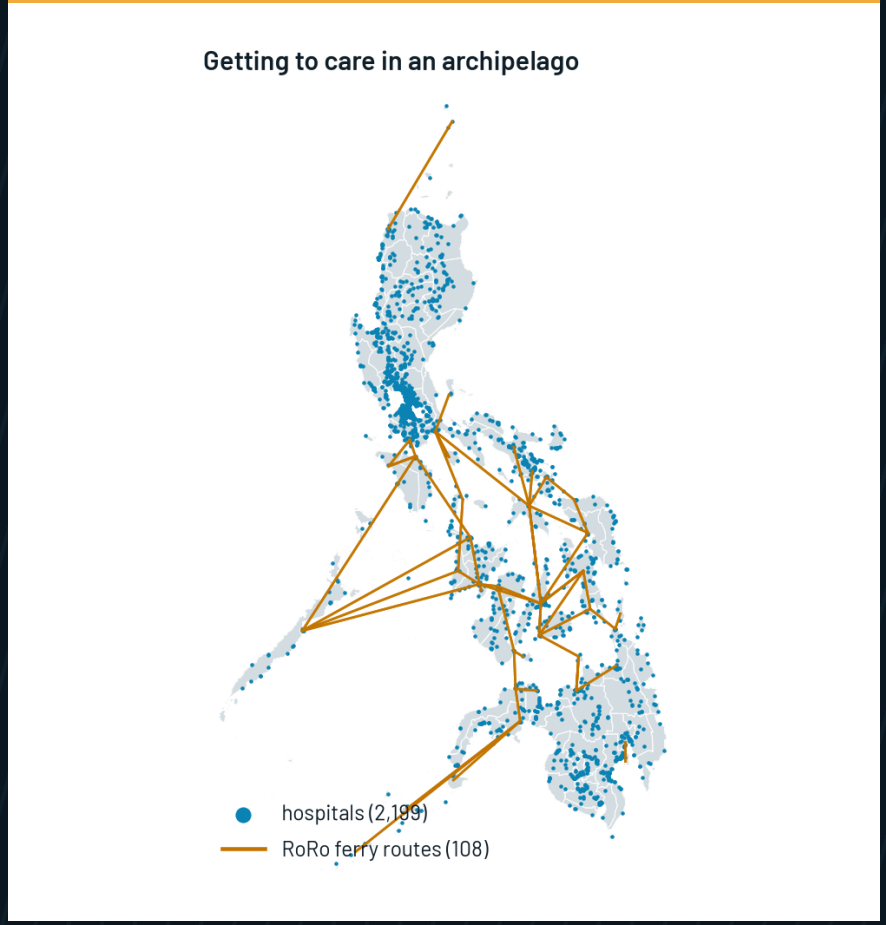
OUTBREAK WAVES ARE KNOWN
Dengue travels between regions (van Panhuis et al., 2015), but this is unused for storm response

AN ARCHIPELAGO MAKES ACCESS THE PROBLEM

- **An island nation.** Many provinces reach the nearest hospital only by RoRo ferry; a gale warning cancels the crossing
- **Roads in floodplains.** The road to the provincial hospital often runs exactly where the river overflows
- **Measured once, for one city.** A 15 cm flood leaves about 22% of Metro Manila without access to higher-level care (Tariverdi et al., 2023); nothing comparable exists nationwide, or with ferries

GUIDING QUESTION

Can flood exposure, a graph of outbreak spread and a storm-disrupted transport network, combined with calibrated forecasts, rank the provinces that need help first, early enough to act?



THREE GRAPHS, ONE RANKING

1 / FLOOD EXPOSURE

Sentinel-1 observed flood, CHIRPS + IMERG rain, rain routed down rivers, UP NOAH hazard, typhoon gale swaths

2 / LEAD-LAG GRAPH

PCMCI+ finds which provinces rise first; their recent rise feeds every follower's forecast

QUANTILE FORECAST

pooled XGBoost, 23 quantiles, 1-4 weeks ahead, tested by rolling origin and scored by WIS

R / SURGE RISK

chance the next 4 weeks beat the province's usual bad month

3 / ACCESS GRAPH

OSM roads + 108 RoRo ferry routes + hospitals; a storm cancels ferries and cuts flooded roads

A / ACCESS LOSS

share of people who lose their route to a hospital

P = R × A

ranked provinces, drivers and an action tier for a real or shifted track

Each half is validated on its own: forecast vs a seasonal baseline; access vs Metro Manila and Copernicus.

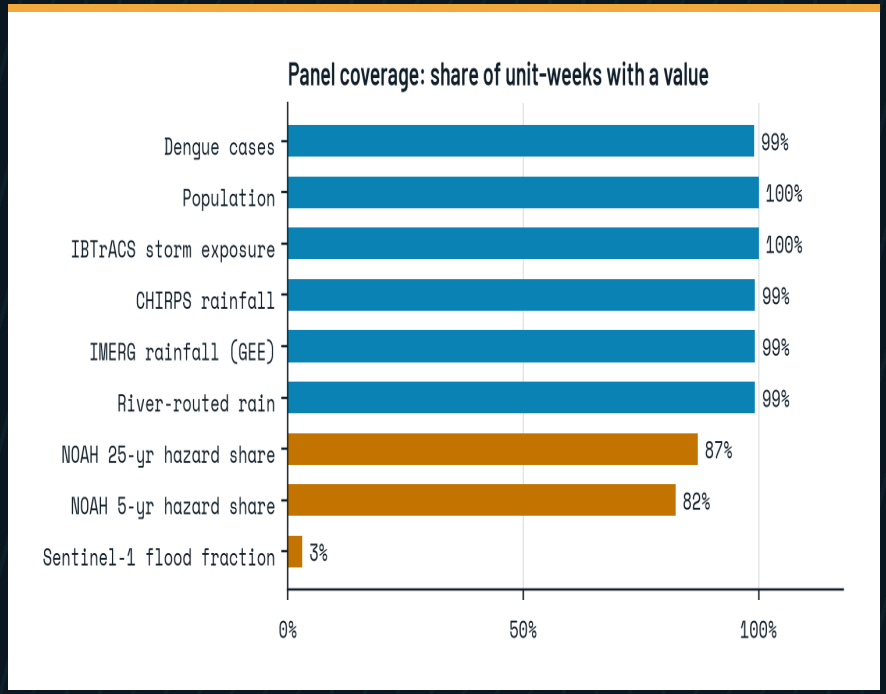
501 OPEN FILES, 22 SOURCES

LAYER	SOURCES
Disease	DOH-Epi dengue, province/city x week, 2016-2021 (HDX); Project CCHAIN; OpenDengue
People and places	2020 census; COD-AB boundaries; WorldPop 2020 100 m; Meta HRSL under-5 / 60+
Rain and rivers	CHIRPS v2.0; GPM IMERG V07; HydroRIVERS
Storms and flood	IBTrACS; Sentinel-1 radar (Google Earth Engine); UP NOAH hazard maps; Copernicus EMS
Getting to care	OpenStreetMap roads; MARINA RoRo routes (roroph); OSM and NOAH health facilities; MAP friction surface

22,270
province-weeks (85 x 262)

2.1 M
road segments, 383,000 km

369k
1 km population cells, 109 M people



Only aggregated, non-identifiable health data.

THE DATA FOUGHT BACK. HOW WE HANDLED IT

WHAT WE FOUND	WHAT WE DID
126 dengue reporting units, not 81 provinces: cities mixed in, misspellings, boundaries changed since 2020	A reviewed crosswalk to 85 harmonised 2016-2020 provinces; a week counts only if every member reported
2020 reporting collapsed under COVID-19 (91k vs 442k cases)	2020 is never a forecast target; it is used for access scenarios only
Flood-hazard maps missing for 11 provinces	Carried as missing, never as zero hazard
Storm wind radii missing on up to 58% of track points	Fallback chain (US, then Japan radius, then fixed), tested in sensitivity
Satellite flood maps disagreed with Copernicus at first	Same-orbit change detection with a dB-difference rule, validated against Copernicus EMS
Short series make causal-discovery algorithms over-confident	False discovery rate control, tested on pure noise first

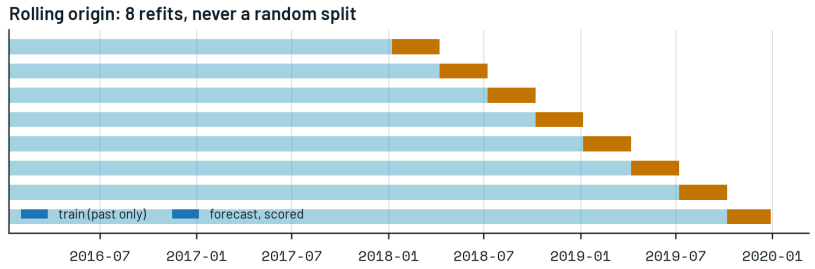
HONEST RANGES, TESTED LIKE REAL TIME

MODEL

- One pooled XGBoost per horizon, 1-4 weeks ahead
- 23 quantiles (the CDC FluSight standard), so every forecast is a range
- Inputs: recent cases, season, rain, river, storm, flood, leader provinces
- A direct 4-week model gives the surge probability R; SHAP explains each one

EVALUATION WITHOUT LEAKAGE

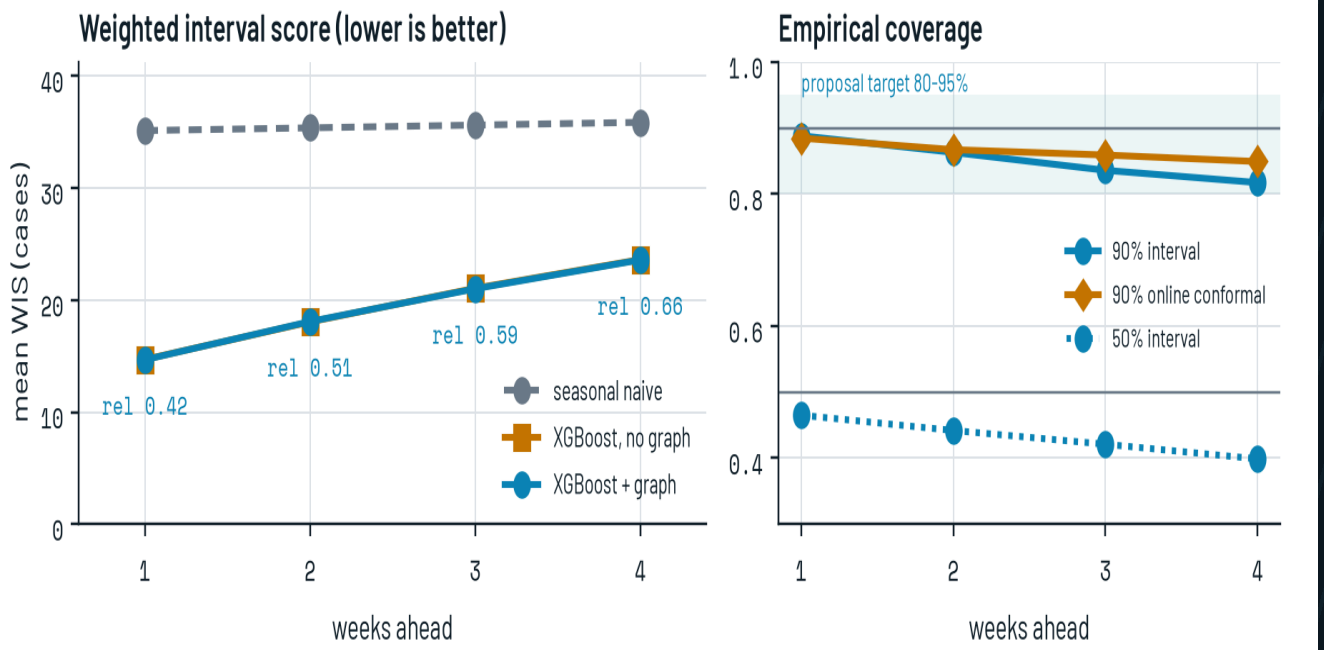
- Rolling origin over 2018-2019: refit every 13 weeks on past data only
- The outbreak graph and surge thresholds are re-learned inside each window
- Weighted interval score on case counts (Bracher et al., 2021)
- Bar to beat: "same week last year" (seasonal naive)



34,050

out-of-sample forecasts scored: 85 provinces, 4 horizons, 2018-2019

46% BETTER THAN THE BAR: RELATIVE WIS 0.54



0.54

relative WIS vs seasonal naive (target below 1): 0.42 one week ahead, 0.66 four weeks ahead

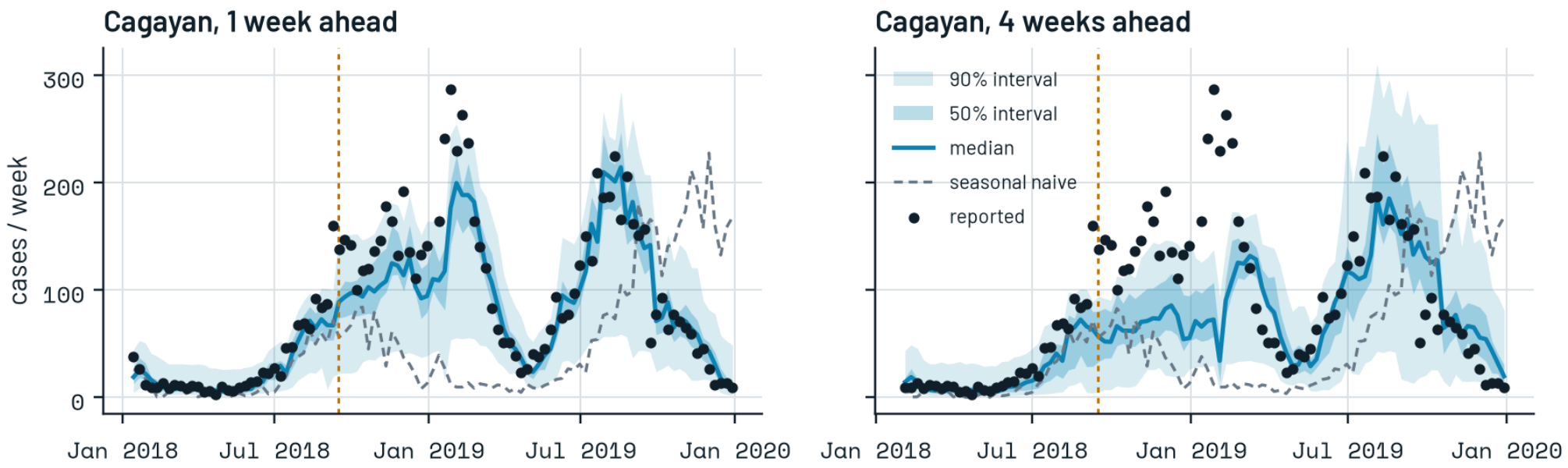
85%

of outcomes fall inside the 90% range (target 80-95%), at every horizon

34k

forecasts scored out of sample, 85 provinces, 2018-2019

WHAT THE FORECAST LOOKS LIKE IN A STORM YEAR



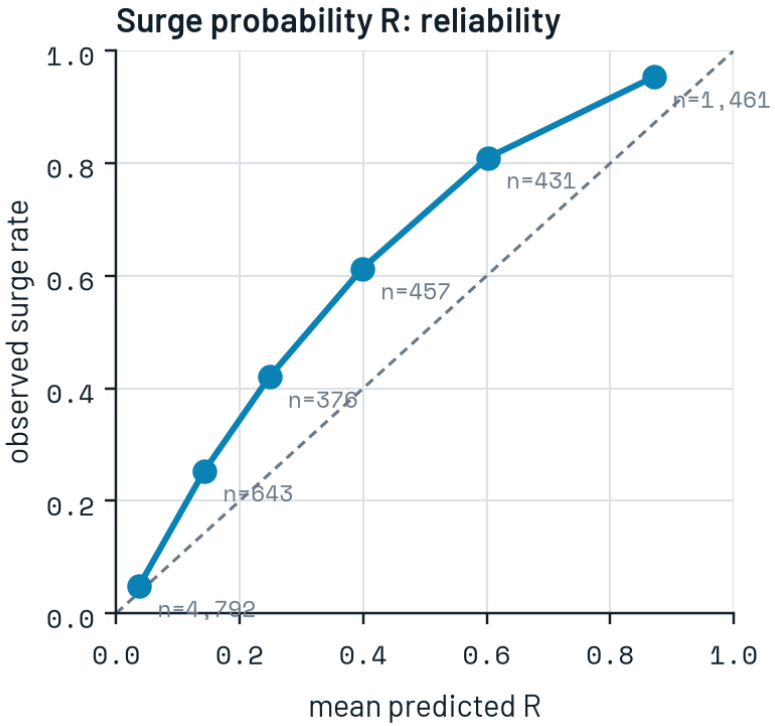
Cagayan, where Ompong made landfall (dashed line). The model is refit every quarter on past data only and still follows the 2018 season and the 2019 epidemic, with ranges that widen at four weeks. The seasonal-naive line (grey) misses the epidemic's timing and size.

SURGE PROBABILITY R: IT RANKS PROVINCES WELL

0.089

Brier score of R, against 0.216 for always predicting the base rate

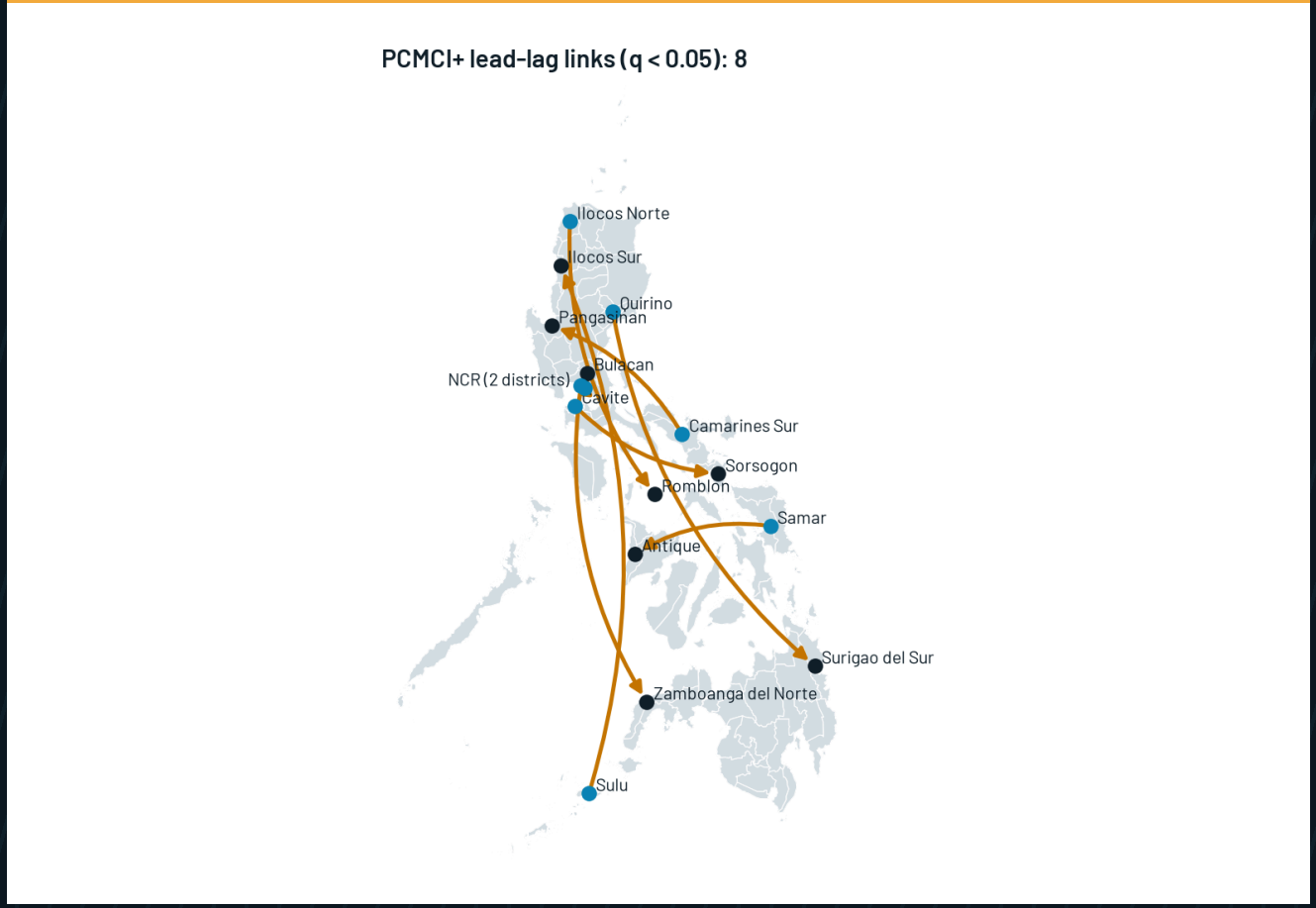
- When R says high, surges follow: above 0.7, 95% of province-weeks surged
- In the middle range R under-states the risk (forecasts of 0.1-0.7 verify 10-20 points higher), so mid-range provinces deserve attention
- 2019's epidemic made surges common (32% of province-weeks)



HOW OUTBREAKS TRAVEL BETWEEN PROVINCES

- PCMCI+ separates "A rises, then B" from "both rise in the rainy season" (Runge, 2020)
- **8 lead-lag links** survive false-discovery control; plain correlation would have drawn 4,265
- Leaders are marked as **sentinel sites** on the dashboard: places to watch first

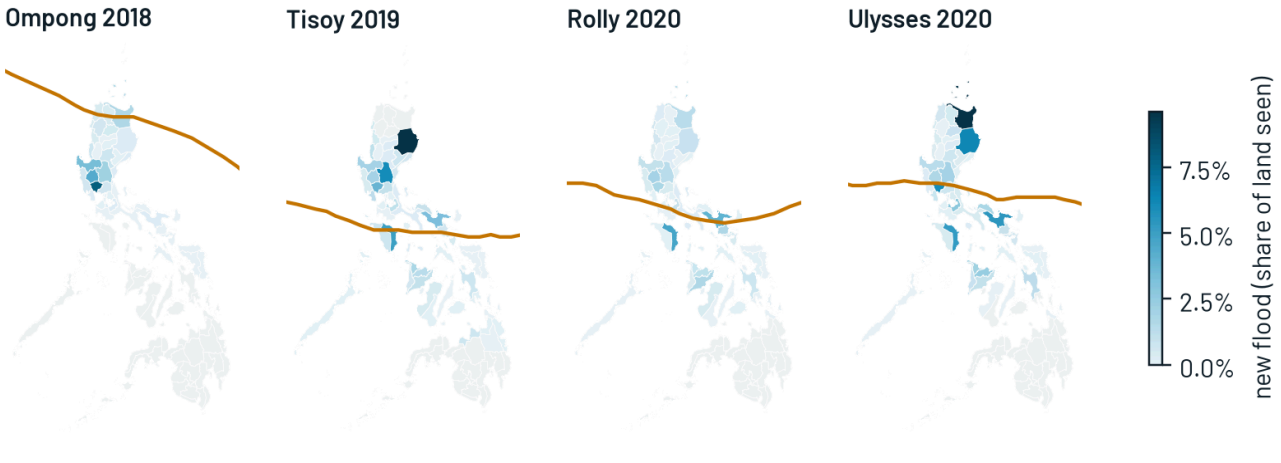
LEADER	FOLLOWER	LEAD	BOOTSTRAP
Cavite	Sorsogon	1 wk	70%
NCR Second	Bulacan	3 wk	70%
Quirino	Surigao del Sur	1 wk	65%
Ilocos Norte	Romblon	1 wk	75%
NCR Third	Zamboanga del Norte	1 wk	85%



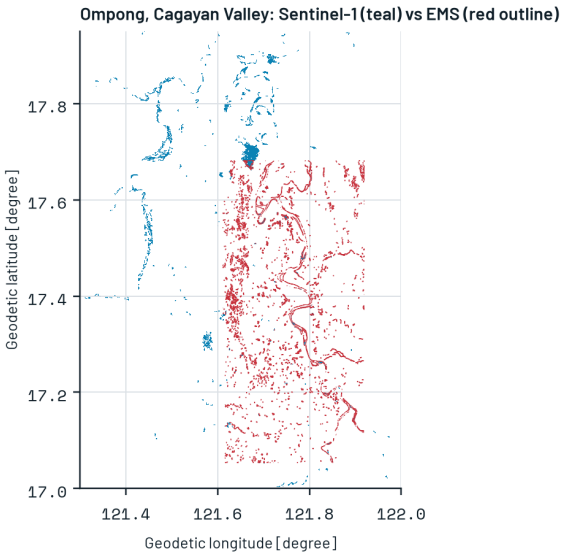
SIGNAL 13 / RESULT 5

DATA: OBSERVED FLOOD EXPOSURE

FLOOD SEEN FROM SPACE, THROUGH THE CLOUDS

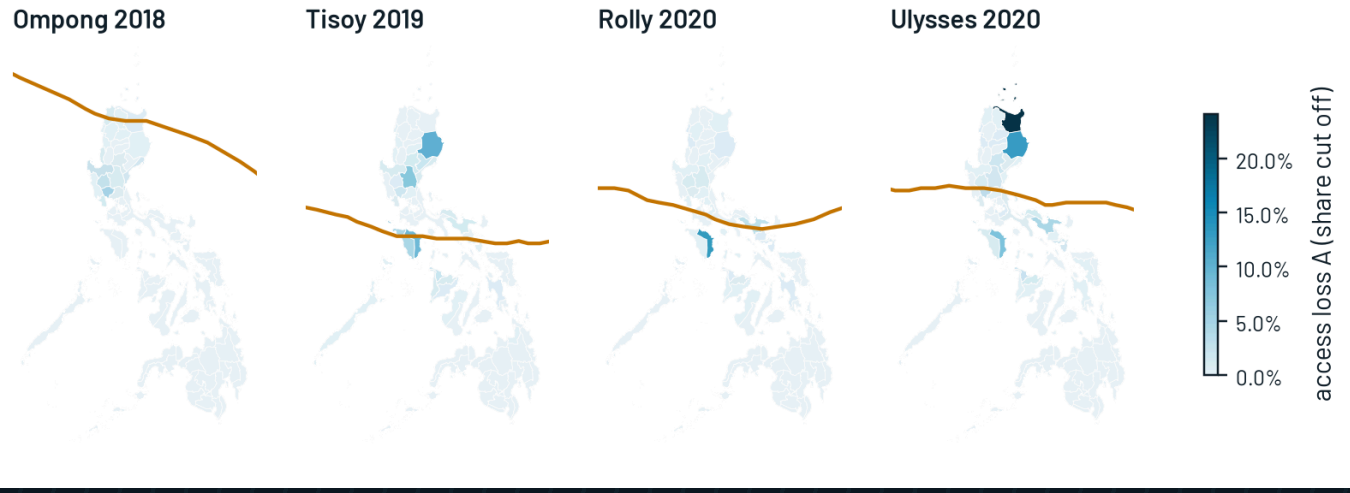


TYPHOON	PEOPLE IN NEW FLOOD	PROVINCES >1% FLOODED	MOST PEOPLE FLOODED
Ompong	0.5 M	5	Pampanga
Tisoy	1.1 M	14	Isabela
Rolly	0.6 M	10	Camarines Sur
Ulysses	1.4 M	14	Cagayan



Checked against Copernicus EMS (Ompong): 55% of Sentinel-1 flood inside the EMS areas matches EMS polygons; Sentinel-1 recovers 24% of the EMS-mapped area (one radar pass every 6-12 days misses short floods).

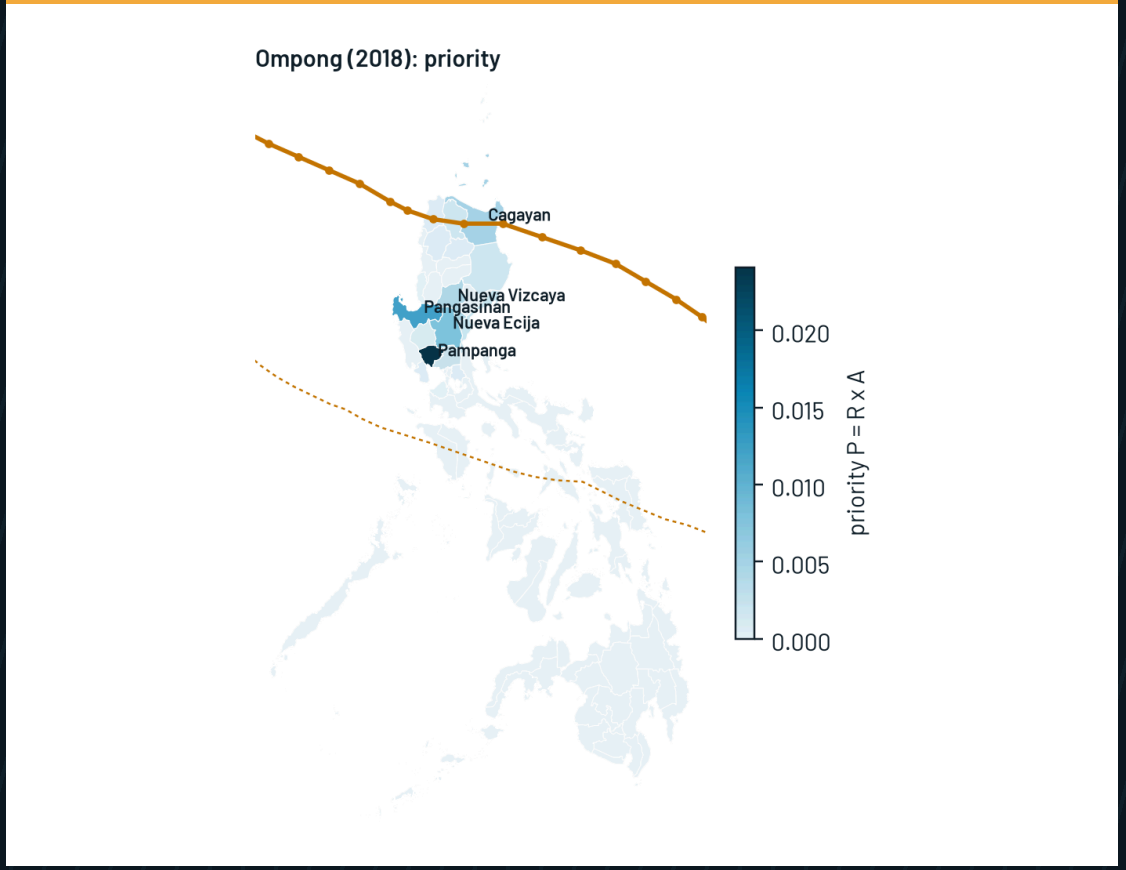
WHO GETS CUT OFF: FOUR TYPHOONS

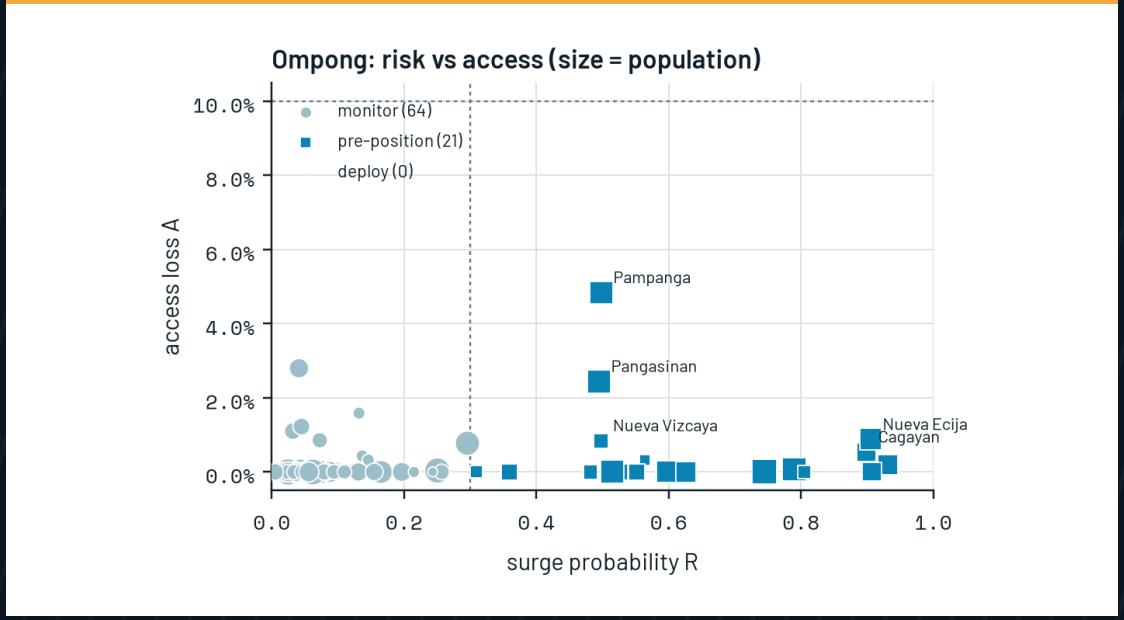


TYPHOON	FERRY ROUTES CUT	PEOPLE CUT OFF	MEAN A	MEAN R	DEPLOY / PRE-POSITION	TOP PRIORITY
Ompong Mangkhut 2018	42	0.3 M	0.2%	0.22	0 / 21	Pampanga
Tisoy Kammuri 2019	100	0.7 M	0.6%	0.08	0 / 5	Camarines Norte
Rolly Goni 2020	42	0.4 M	0.3%	0.02	0 / 0	Oriental Mindoro
Ulysses Vamco 2020	54	1.0 M	0.8%	0.04	0 / 2	Cagayan

Roads cut only where radar saw new water: a lower bound. Using flood-hazard maps instead gives 4.8-10.3 M people cut off: an upper bound.

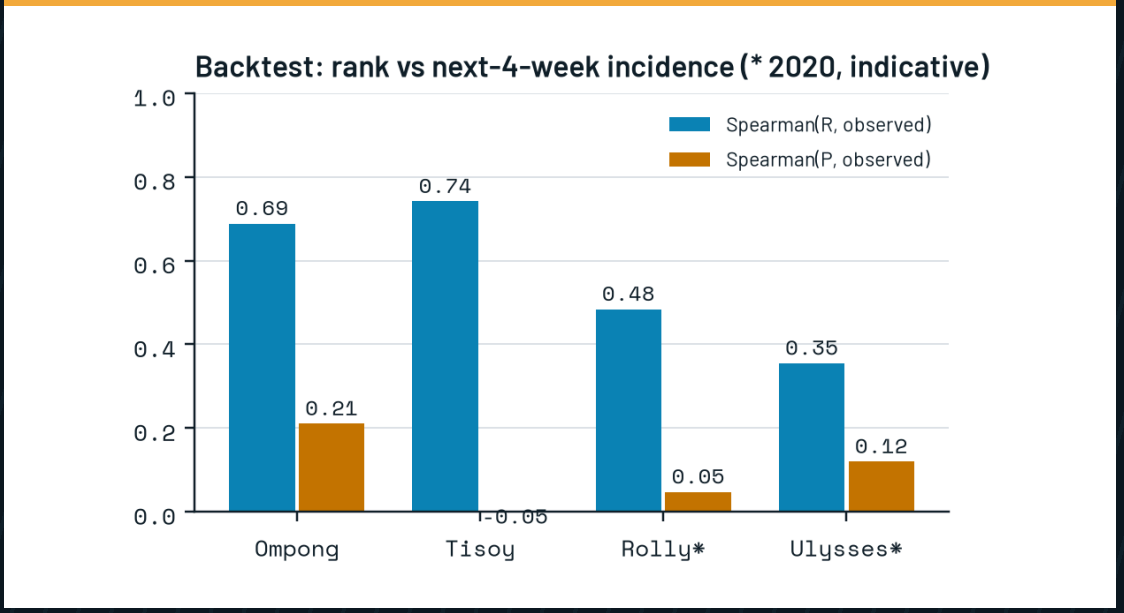
OMPONG 2018: PAMPANGA FIRST





Using data up to the landfall week only: 21 provinces flagged to **pre-position supplies** before the surge. In the next four weeks, **7 of the top 10** really surged.

IT POINTS AT THE RIGHT PLACES



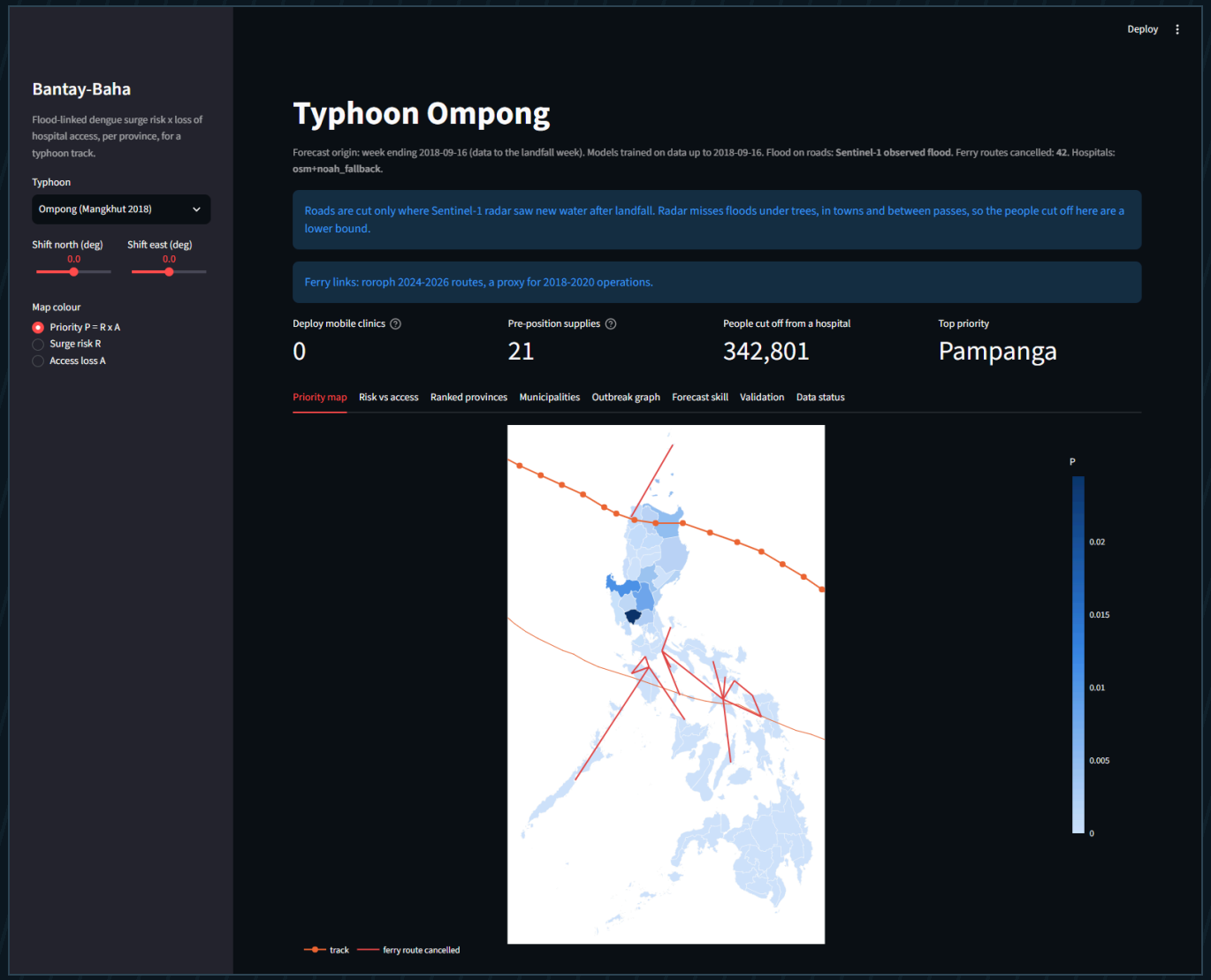
- **Backtest**: R's ranking matches the next four weeks' incidence, ρ = 0.69 (Ompong) and 0.74 (Tisoy)
- **Ompong**: 7 of the 10 highest-priority provinces surged, against a 40% base rate
- **Robust**: nine changes to travel speeds, ferry rules, cut-off rules and storm radius leave the ranking in place (Kendall's τ ≥ 0.99)
- **Benchmark**: 30% of Metro Manila cut off when hazard-zone roads close; Tariverdi et al. (2023) found ~22% for a 15 cm flood, the same order

0.74
rank correlation, Tisoy

≧0.99
ranking stability (Kendall's τ)

30%
NCR cut off vs ~22% published

DEMONSTRATION VIDEO: FUNCTIONAL SHOWCASE



WHAT A HEALTH OFFICER SEES

- Pick a typhoon, or shift its track north, south or east
- A priority map with the track, gale swath and cancelled ferries
- Risk vs access, with the action tier for every province
- Expected cases for the next 4 weeks with a 90% range, people cut off, and the drivers behind each risk
- Drill down to municipalities
- Answers in seconds: every scenario is pre-computed

PUTTING IT IN THE LOOP

1
STORM WATCH

PAGASA issues a tropical cyclone bulletin. Its forecast track goes in, along with shifted tracks for the cone of uncertainty.

2
PRE-LANDFALL BULLETIN

Within minutes: a ranked province list with action tiers to DOH regional offices, provincial health offices and DRRMOs, 48-72 h ahead.

3
POST-LANDFALL UPDATE

Days 1-7: Sentinel-1 flood maps replace the hazard estimate; ferry and road status updates who is cut off.

4
LEARN EVERY WEEK

Weeks 1-8: each PIDSR report is compared with the forecast; the model recalibrates and tiers are adjusted.

The output is an **action**, not a chart: monitor sentinel sites, pre-position doxycycline and vector control, or send mobile clinics **before** the roads and ferries close.

DOH REGIONAL OFFICE

Ranked provinces with action tiers and expected cases (90% range) for the next 4 weeks

PROVINCIAL HEALTH OFFICE

Which municipalities lose hospital access, and the drivers behind the risk

DRRMO

Ferry routes and roads expected to close: where mobile clinics must go first

BUILT TO BE USED, NOT JUST SHOWN

WHERE THE USERS ARE

- Web dashboard for regional epidemiologists
- One-page PDF bulletin per storm for provincial health offices
- SMS summary for the "deploy" provinces
- Low bandwidth: maps need no tile server

DESIGNED WITH THEM

- Action tiers co-designed with DOH and LGU health officers and DRRMOs
- Filipino and English labels
- Usability tested (System Usability Scale) before each typhoon season

RUNS ON WHAT EXISTS

- Weekly PIDSR counts, PAGASA tracks, Sentinel-1, the DOH facility registry
- Extends to leptospirosis with city-level surveillance

~4 s
to recompute hospital access for a new track, nationwide

~6 s
for the dashboard to load a scenario

100%
open data and open code

1
command rebuilds everything when new data arrive

The novelty is the join: **storm-specific** disease risk × **storm-specific** loss of access, ferries included, in one ranking a health officer can act on.

SIGNAL 20 / SOURCES

SOURCES: APA 7TH EDITION

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